

Basic Principles of Medicine 2

Module : Digestion and Metabolism

DLA

Autonomics of the Abdomen

St Georges University ©

Objectives

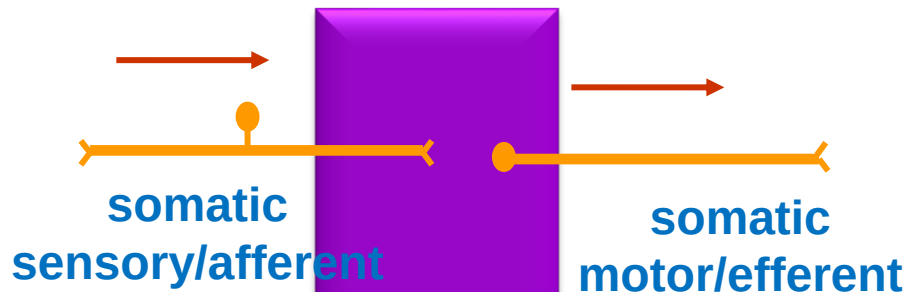
SOM.MK.1.BMP2.3.DM.1.ANAT.1110	Describe the location of the abdominal sympathetic chain.
SOM.MK.1.BMP2.3.DM.1.ANAT.1111	Describe the location of the sympathetic ganglia associated with the abdominal organs.
SOM.MK.1.BMP2.3.DM.1.ANAT.1112	Describe the contributions from the thoracic splanchnic and lumbar splanchnic nerves to the para- and preaortic sympathetic ganglia.
SOM.MK.1.BMP2.3.DM.1.ANAT.1113	Describe the sympathetic, parasympathetic and sensory innervation of the foregut, midgut and hind gut.
SOM.MK.1.BMP2.3.DM.1.ANAT.1114	Describe the effect of sympathetic and parasympathetic innervation on the gastrointestinal system.
SOM.MK.1.BMP2.3.DM.1.ANAT.1115	Describe how pain from the abdominal organs may be referred to the body wall.
SOM.MK.1.BMP2.3.DM.1.ANAT.1116	Describe the location of the referred pain from the gastrointestinal tract and organs based on innervation and embryological origin.

Review: Nervous System Integration

Somatic

Receptors in **BODY WALL** for conscious sensation:

- pain
- pressure
- touch
- temperature
- proprioception



Somatic

Voluntary control of:

- **skeletal muscle**

PNS

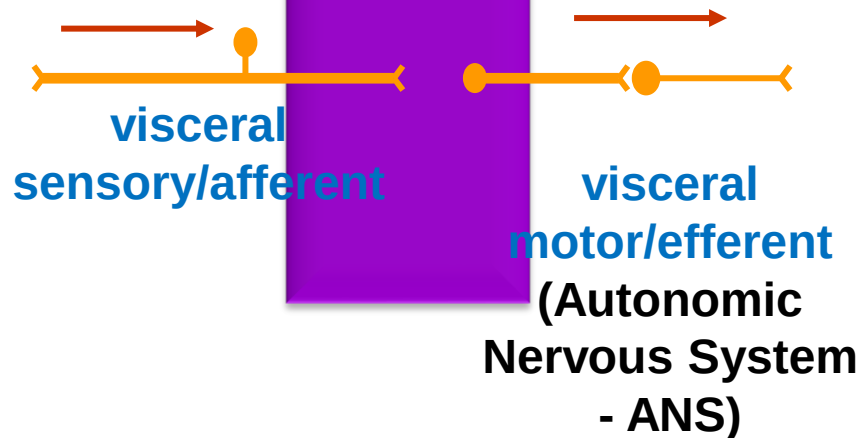
CNS

PNS

Visceral

Receptors in **ORGANS** and **TISSUES**:

- distension
- inflammation
- ischemia
- physiological changes
- abnormal muscle spasm



Visceral

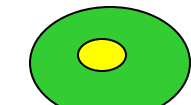
Involuntary control of:

- **smooth muscle**
- **cardiac muscle**
- **glands**

Review: Divisions of the Autonomic Nervous System

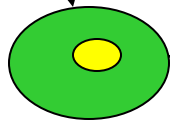
Central Nervous System

Sympathetic



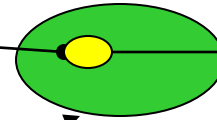
nuclei

Parasympathetic

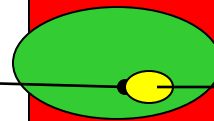


Peripheral Nervous System

ganglion



Target
smooth muscle
glands
cardiac muscle



Target
smooth muscle
glands
cardiac muscle

Preganglionic =
myelinated

Postganglionic =
unmyelinated

Review: Preganglionic

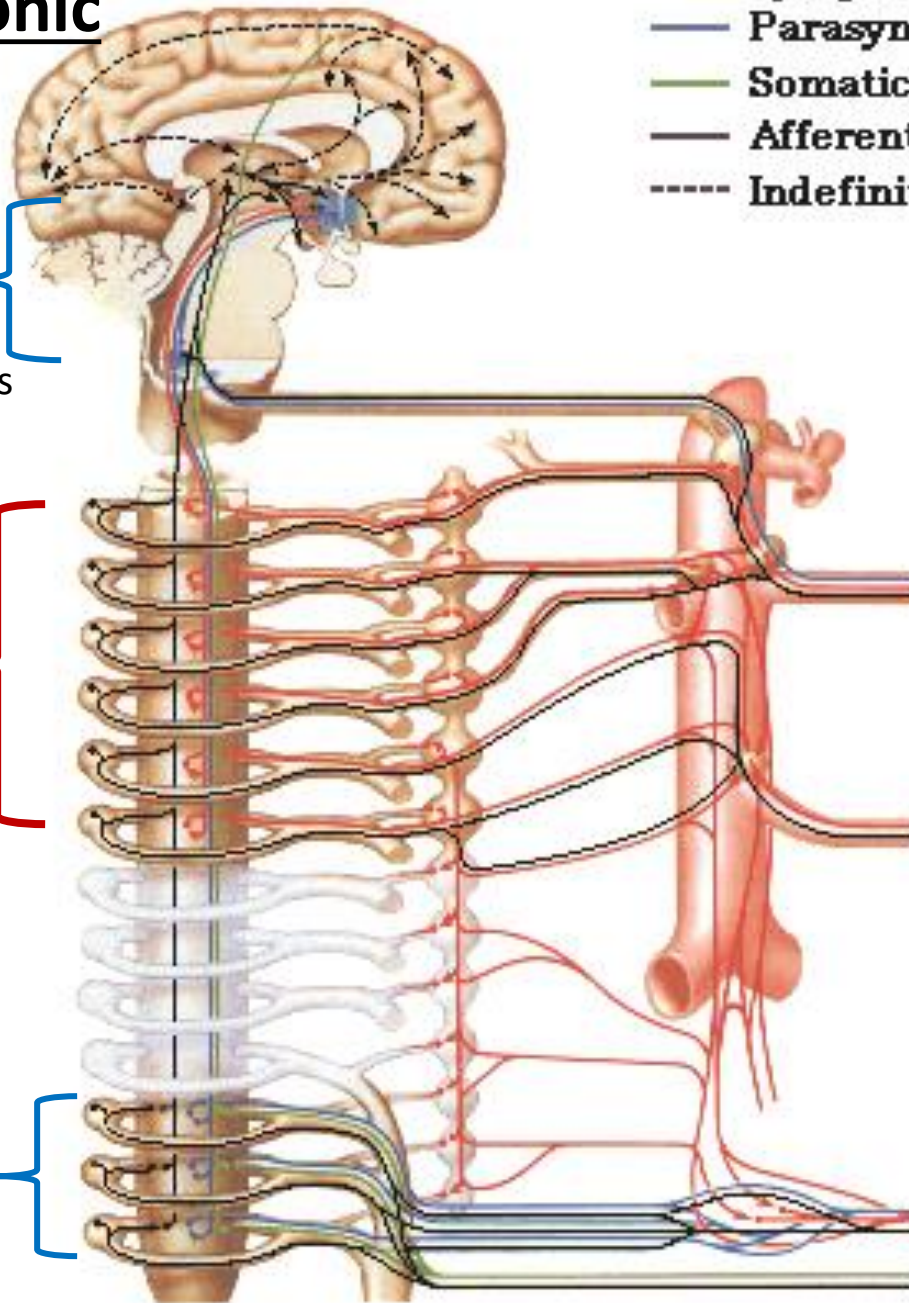
Parasympathetic Outflow

Brainstem: Cranial Nerves
III, VII, IX, X

**Sympathetic
Outflow**
Lateral horn of
Thoracolumbar
spinal cord
T1 –L2

Sacral spinal
segments
S2-S4

- Sympathetic efferents
- Parasympathetic efferents
- Somatic efferents
- Afferents and CNS connect
- Indefinite paths



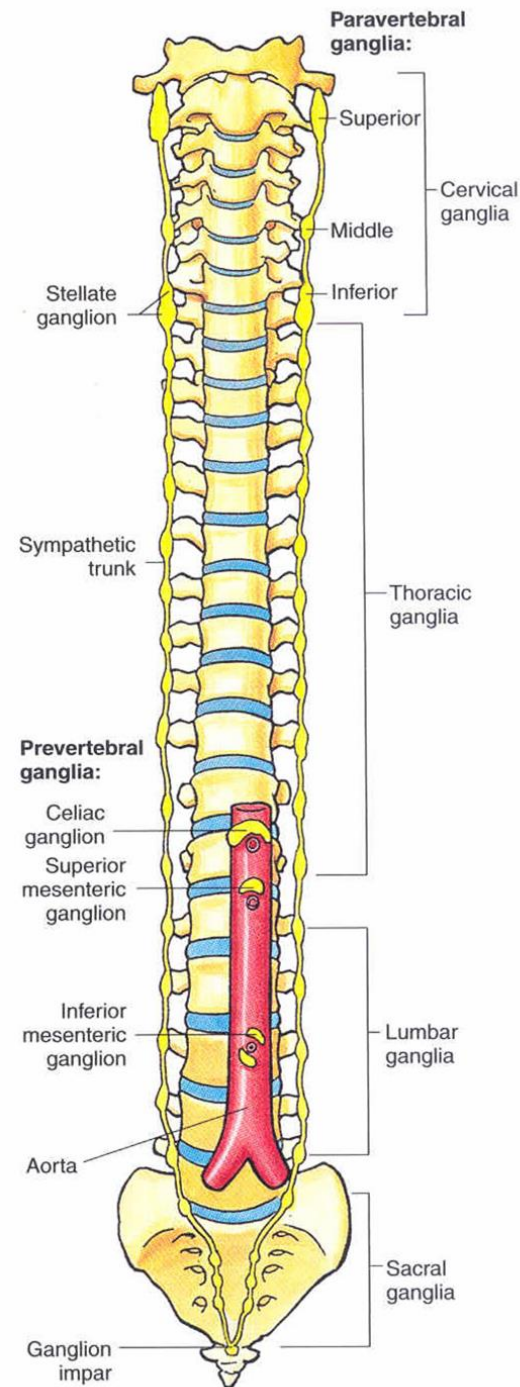
Review: Postganglionic Sympathetic

Paravertebral or chain ganglia

- three in cervical region
- one associated with each ventral ramus in other regions

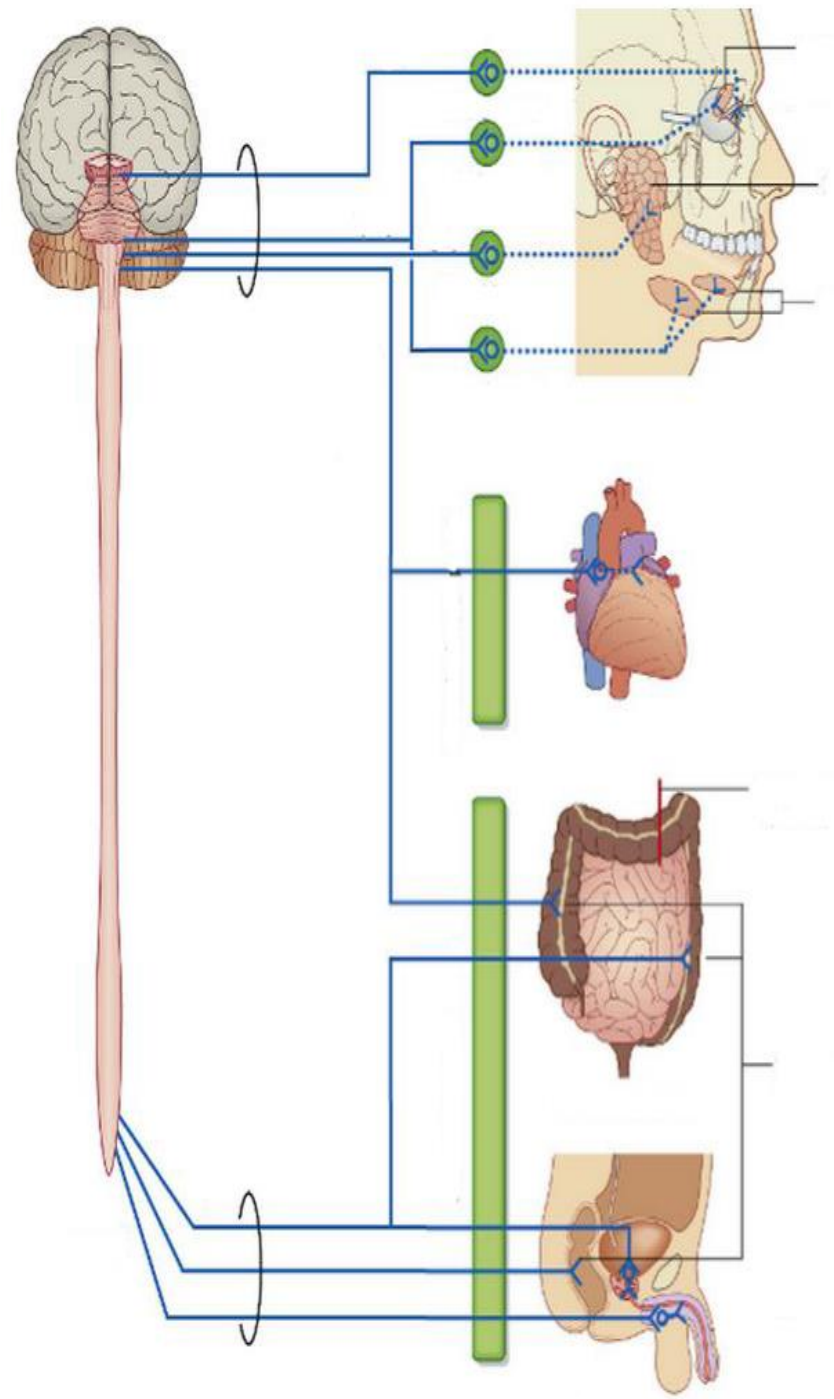
Preaortic/prevertebral ganglia

- at roots of major arteries branching from abdominal aorta
 - Celiac
 - Superior mesenteric
 - Inferior mesenteric
 - Renal



Review: Postganglionic Parasympathetic

- found scattered near or in the walls of target organs
- discrete parasympathetic ganglia are only found in the head associated with the parasympathetic cranial nerves
- In the GI: synapse on the neurons of the submucosal (Meissner) and myenteric (Auerbach) plexus (*Enteric system*)



Functions of the ANS in the GI

Sympathetic:

- Vasoconstriction
- Constriction of sphincters

Parasympathetic:

- Increase GI motility
- Increase GI secretion

Enteric:

- *Intrinsic neuronal circuit consisting of motor and sensory neurons organized in two interconnected plexuses (myenteric and Meissner's)*
- *Generally independent, but receives input from both the parasympathetic and sympathetic systems*

Derivatives of the Foregut

Pharynx and its derivatives,
Lower respiratory tract,
Esophagus, Stomach
Duodenum up to the major duodenal papilla
Liver, Biliary apparatus and Pancreas

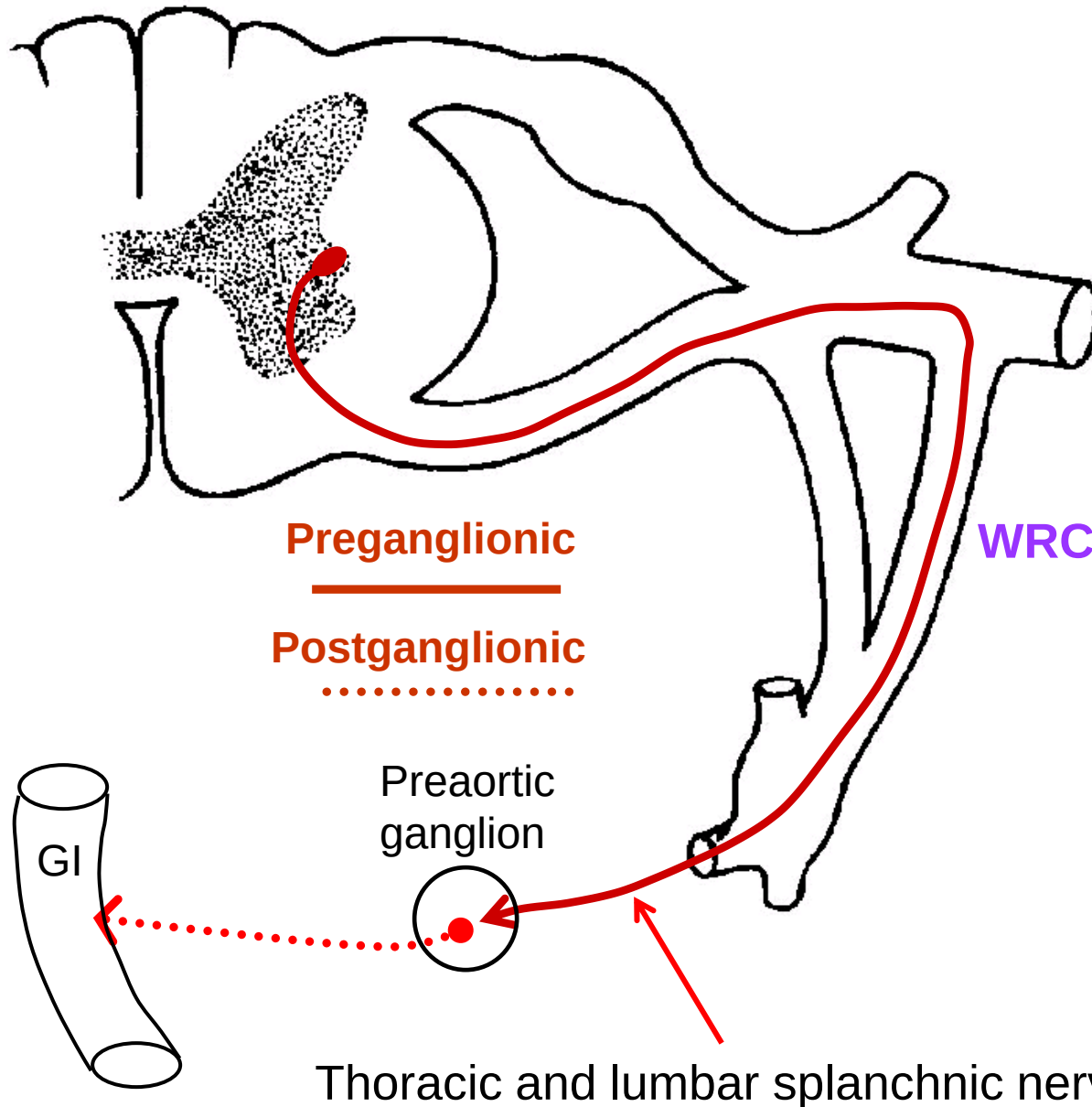
Derivatives of the Midgut

Duodenum distal to the major duodenal papilla,
Jejunum & Ileum; Cecum & Appendix,
Ascending colon & right 2/3 of transverse colon

Derivatives of the Hindgut

Left 1/3 of Transverse colon,
Descending colon, Sigmoid colon & rectum
Anal canal up to the pectinate line
Urinary bladder & most of Urethra

Sympathetics to Gastrointestinal System



- Preganglionic fiber enters chain via WRC
- Passes on through ganglion **WITHOUT** synapsing
- Still preganglionic fiber then leaves the medial aspect of the ganglion and travels to a sympathetic ganglion on the abdominal aorta- PREAORTIC GANGLION
- Postganglionic fibers then travel along plexuses on the arteries to the GI tract

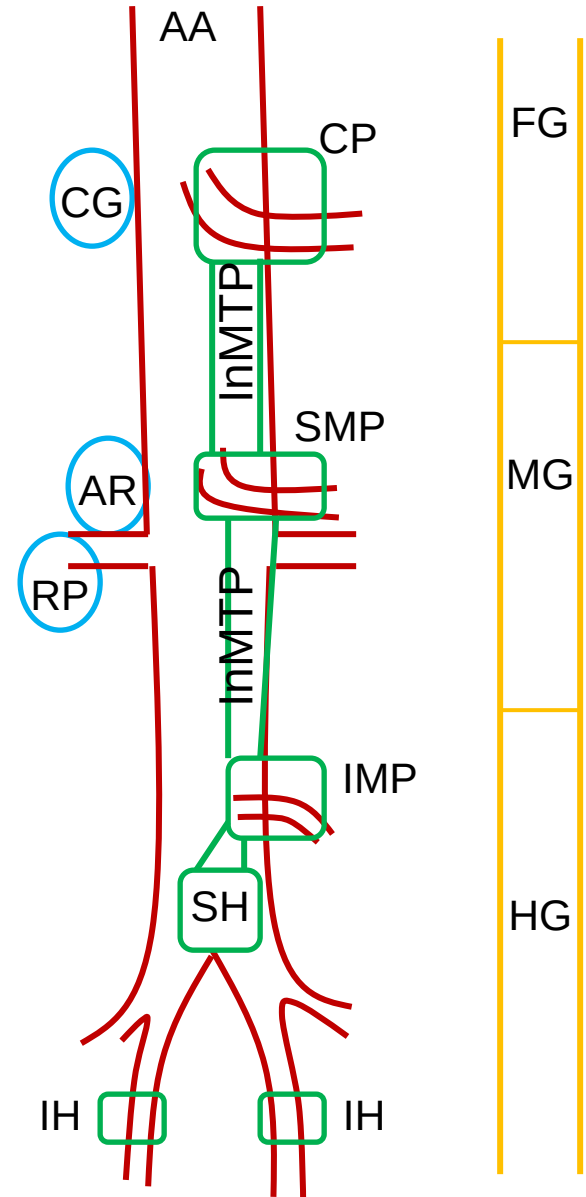
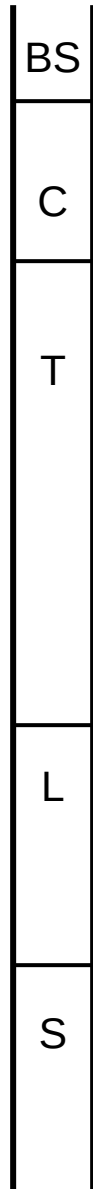
Sympathetic Splanchnics (Preganglionic) to GI

- **Greater Splanchnic(T5-T9)** synapses in the celiac ganglion.
- **Lesser Splanchnic(T10-11)** synapses in the aortico-renal ganglion.
- **Least Splanchnic (T12)** synapses in the renal plexus.
- **Lumbar Splanchnic (L1-L2)** synapses in the intermesenteric and/or superior hypogastric plexus

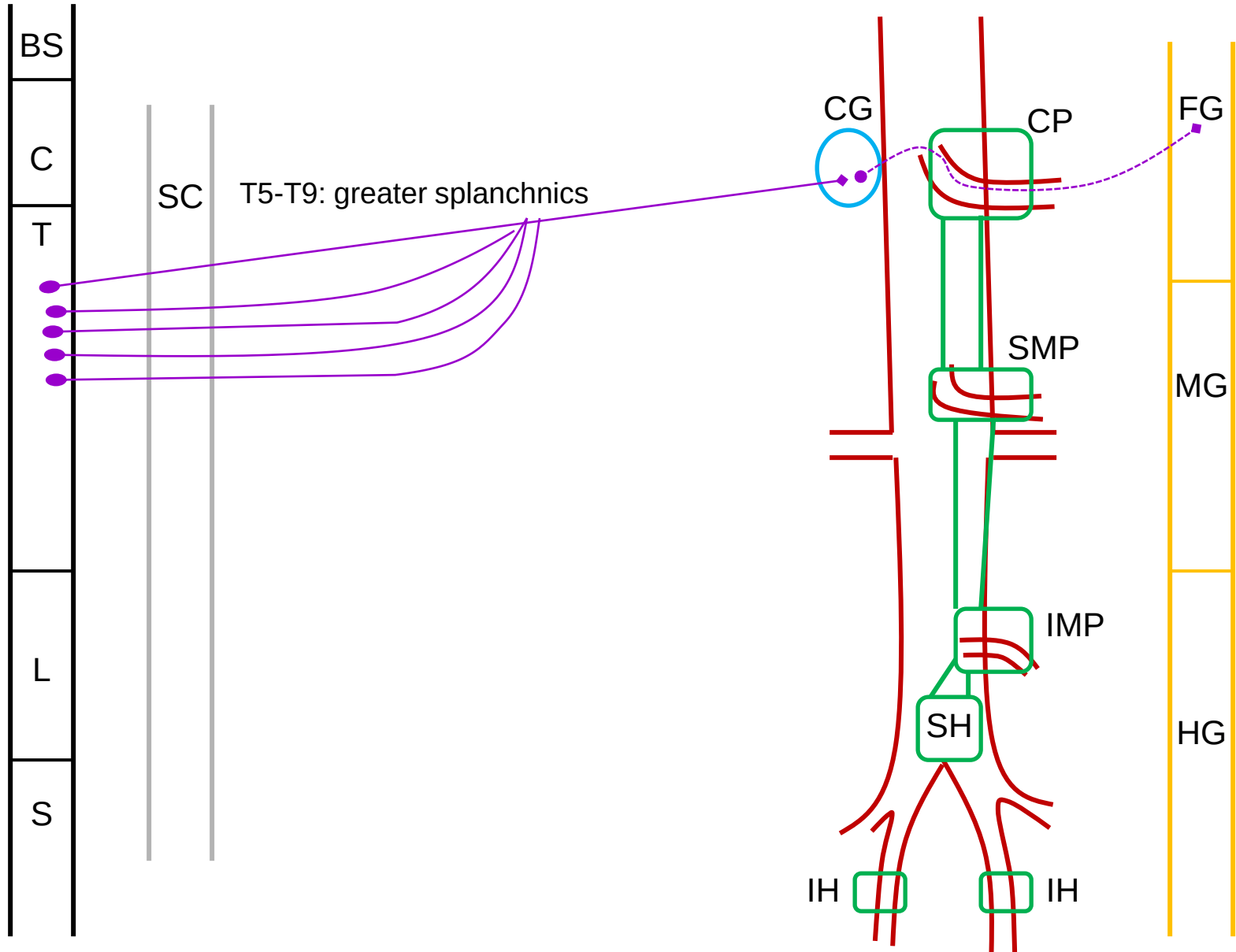
Sympathetic Postganglionic to the GI

- Postganglionic sympathetic fibers travel on **peri-arterial plexuses** to get to their target organs.
- Arterial supplied based on embryonic origin:
 - **Foregut**- Celiac artery
 - **Midgut**- Superior mesenteric artery
 - **Hindgut**- Inferior mesenteric artery

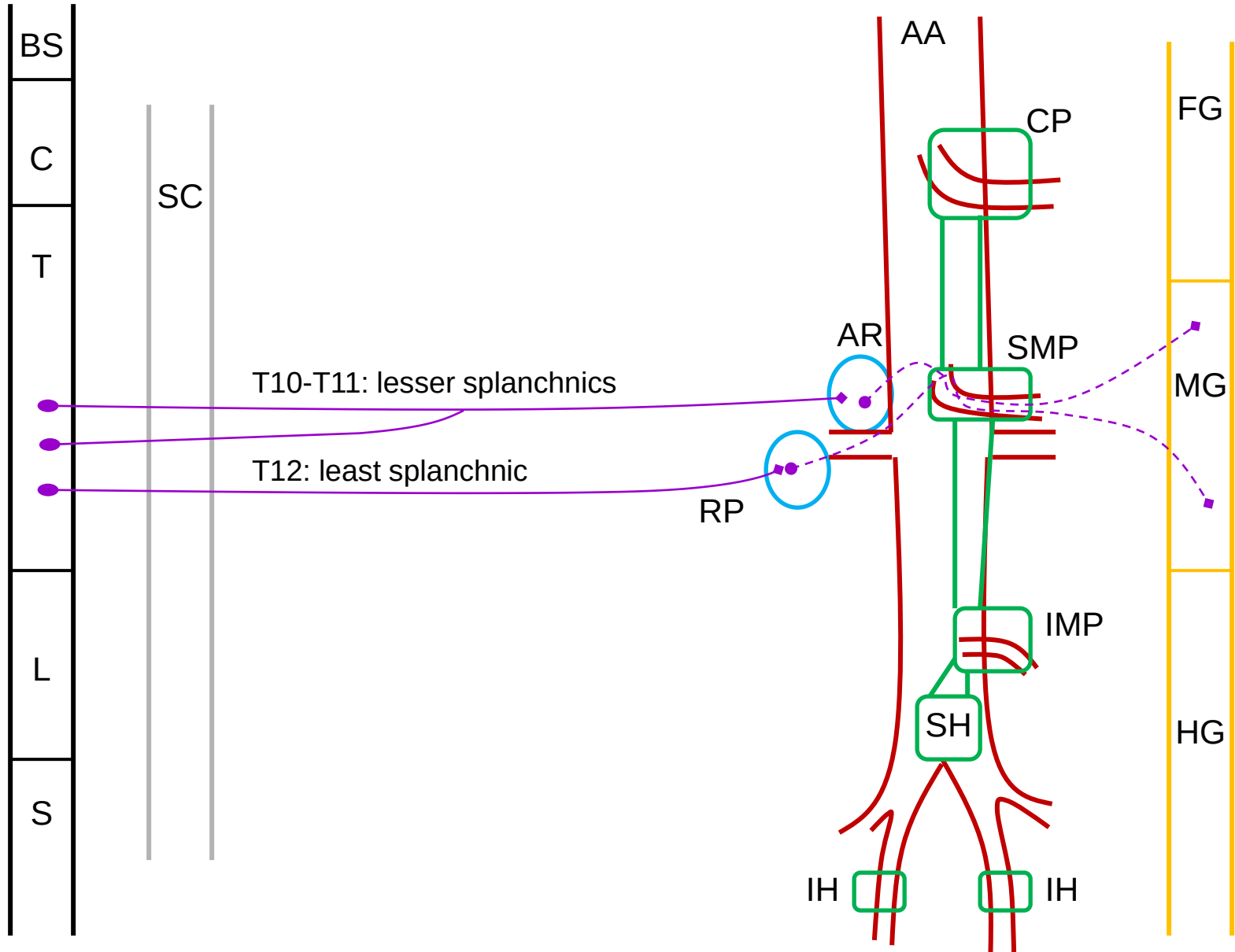
Sympathetics



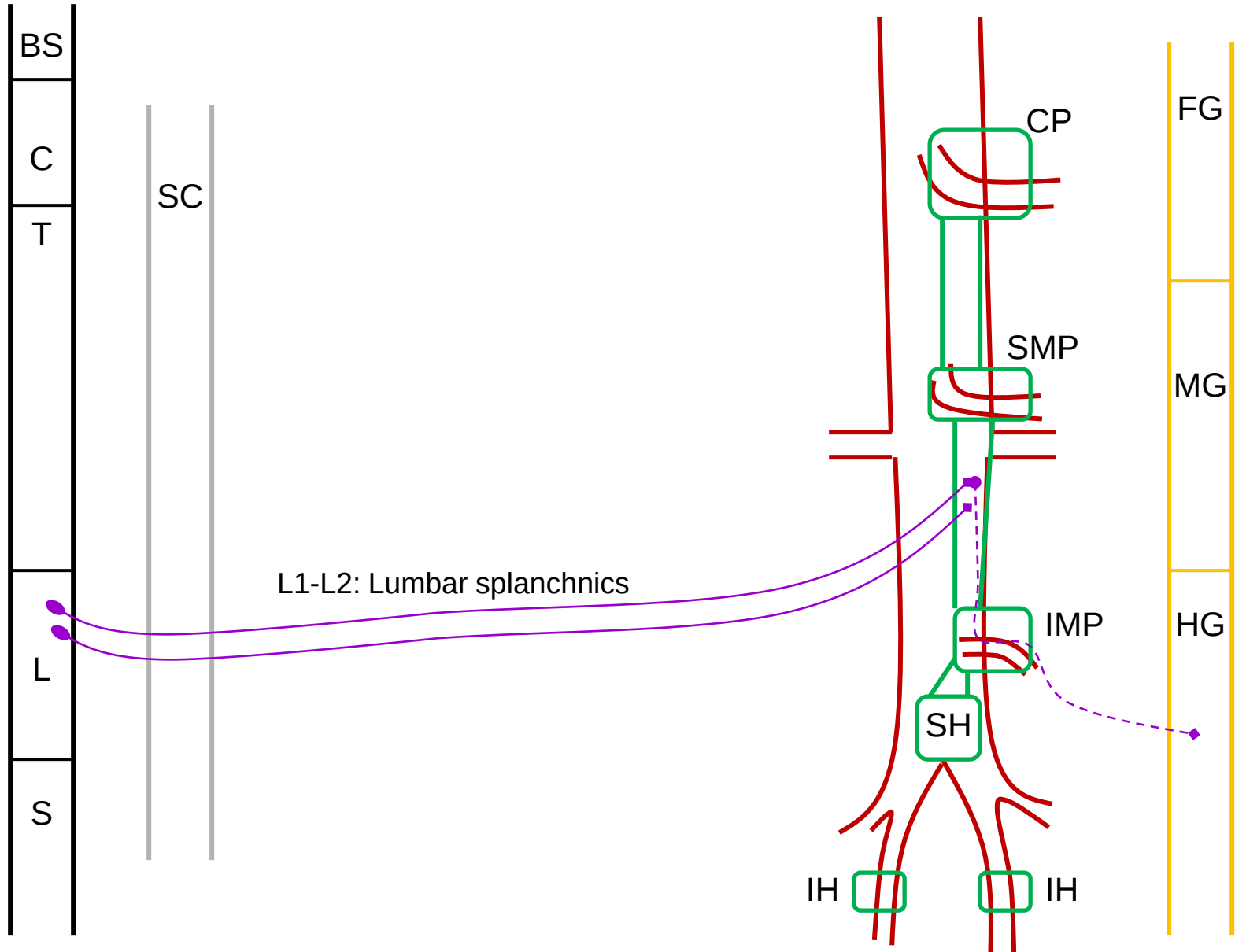
Sympathetics



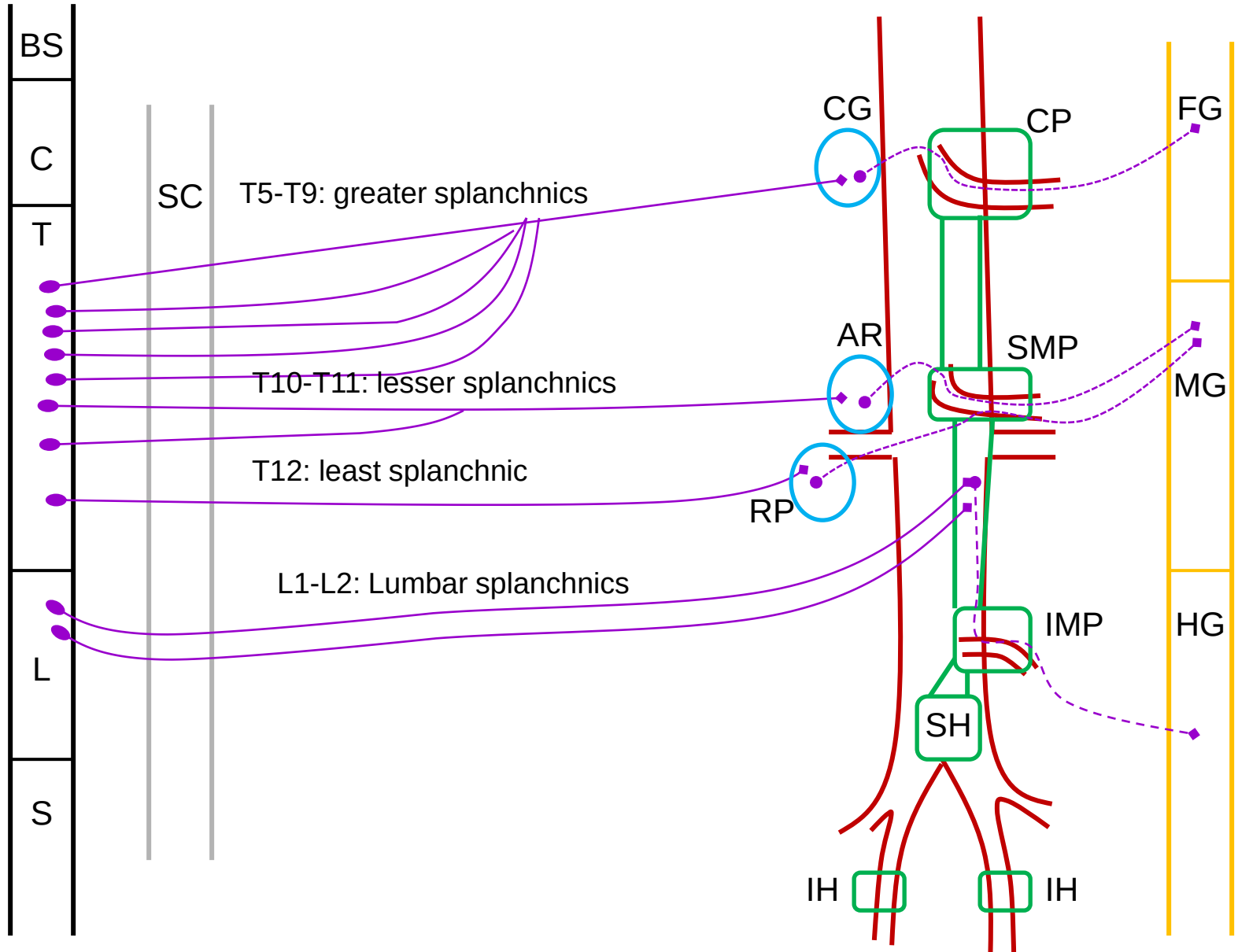
Sympathetics



Sympathetics



Sympathetics



* Note that greater, lesser, least and lumbar splanchnics also provide preganglionic to the kidneys, ureters and pelvic organs

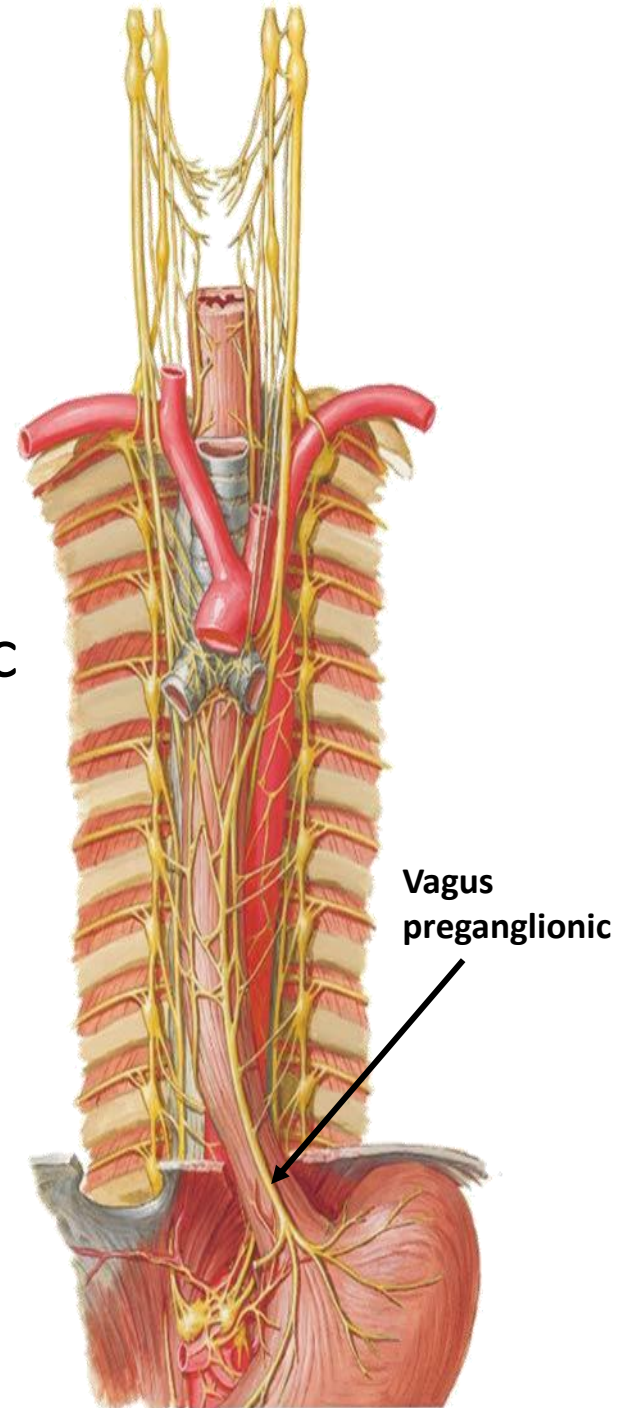
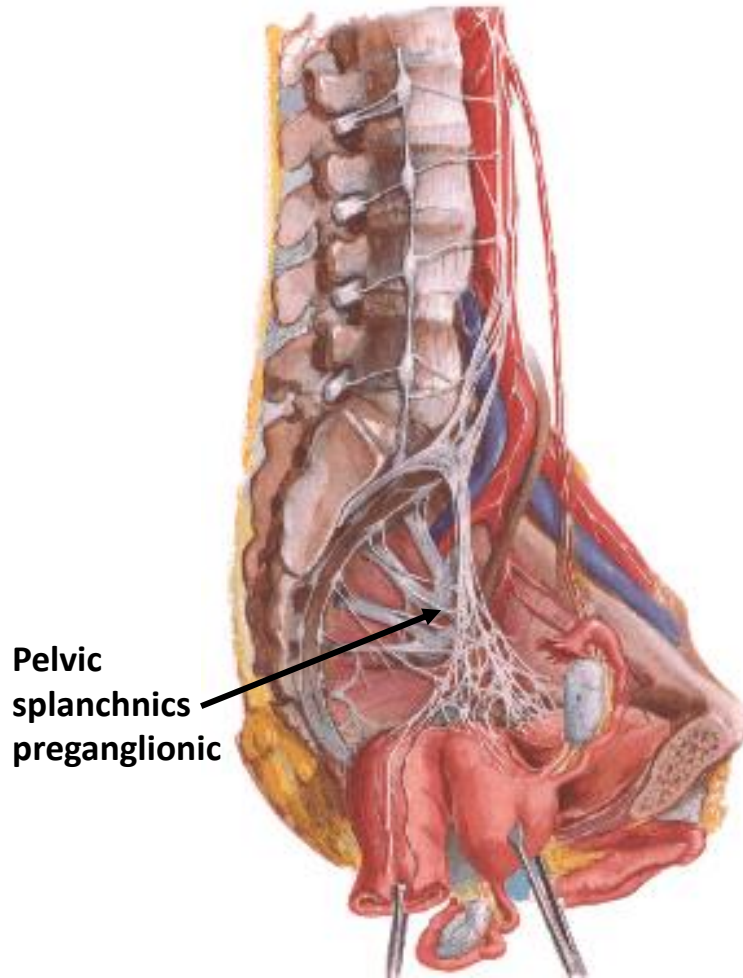
Parasympathetics to Gastrointestinal System

Vagus nerve

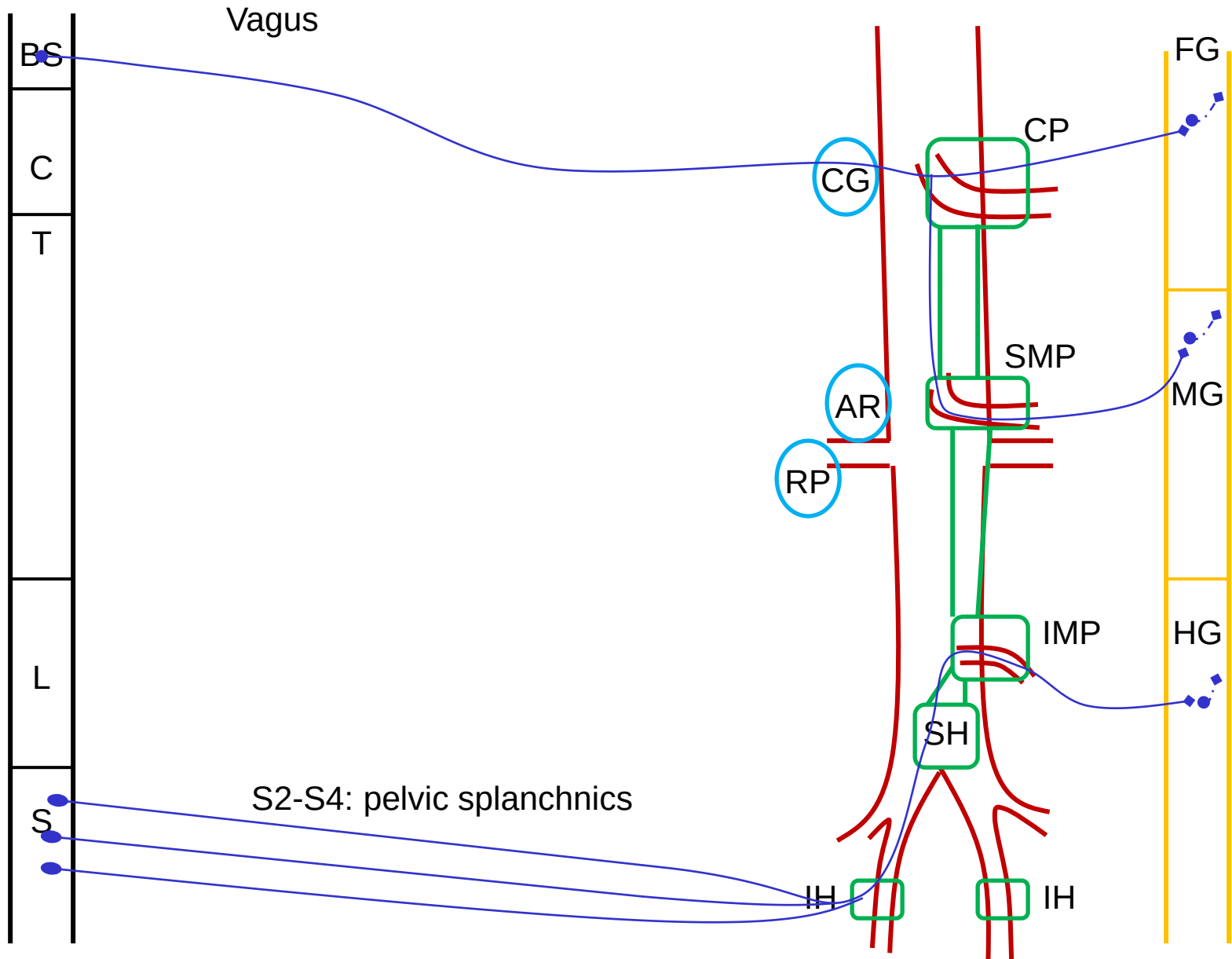
- Foregut and midgut

Pelvic splanchnic S2-S4

- Hindgut



Parasympathetics

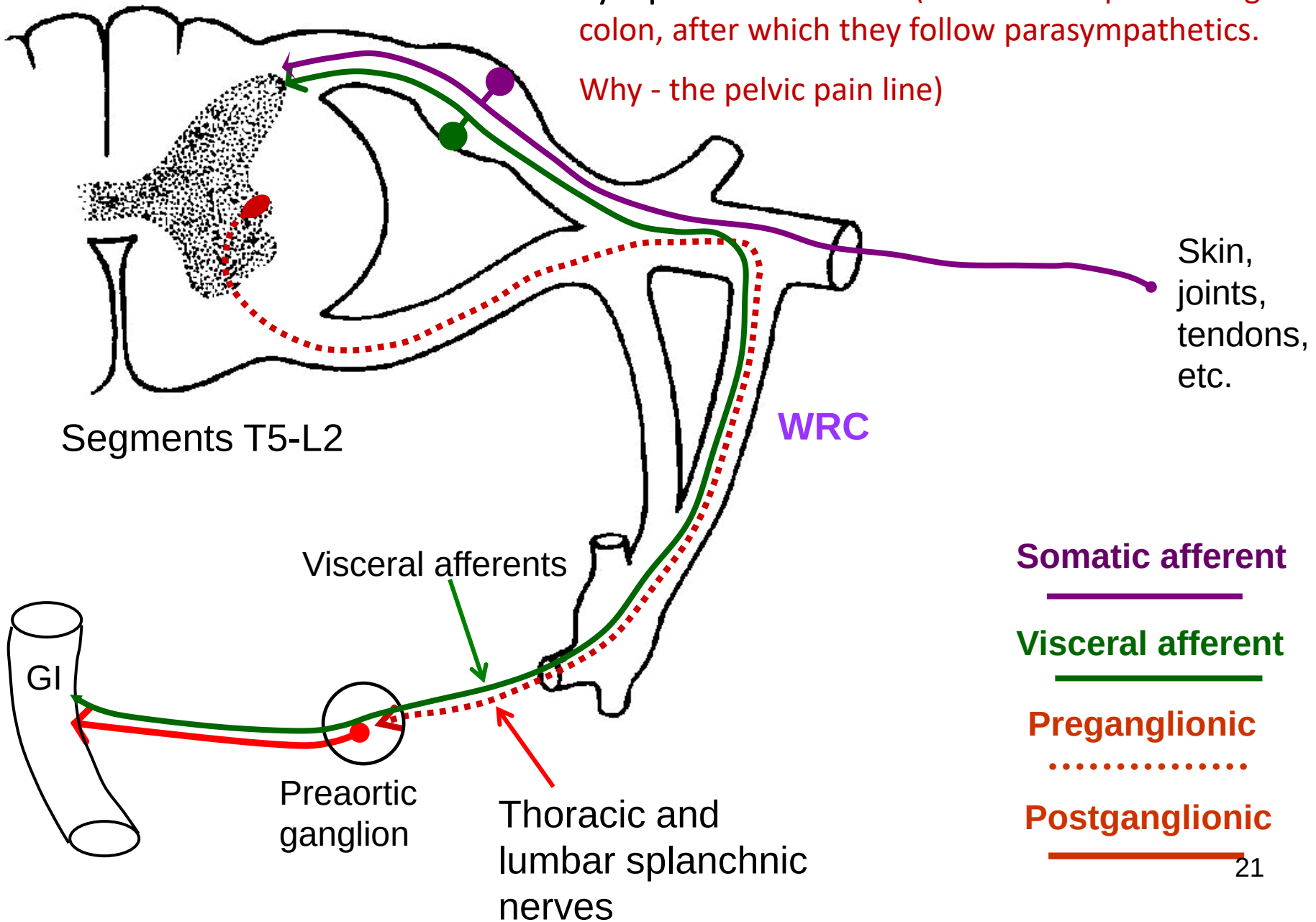


VISCERAL SENSORY FIBERS

- Visceral afferents fibers that mediate pain sensation (e.g. respond to ischemia, inflammation and distention) travel with **sympathetic fibers** until the midpoint of the sigmoid colon where they then follow parasympathetics.
- Therefore by knowing the preganglionic sympathetic supply to a particular organ, you will be able to determine where in the body pain from those organs will refer to.

Visceral sensory fibers hitchhike on sympathetic fibers. (until the midpoint of sigmoid colon, after which they follow parasympathetics.

Why - the pelvic pain line)



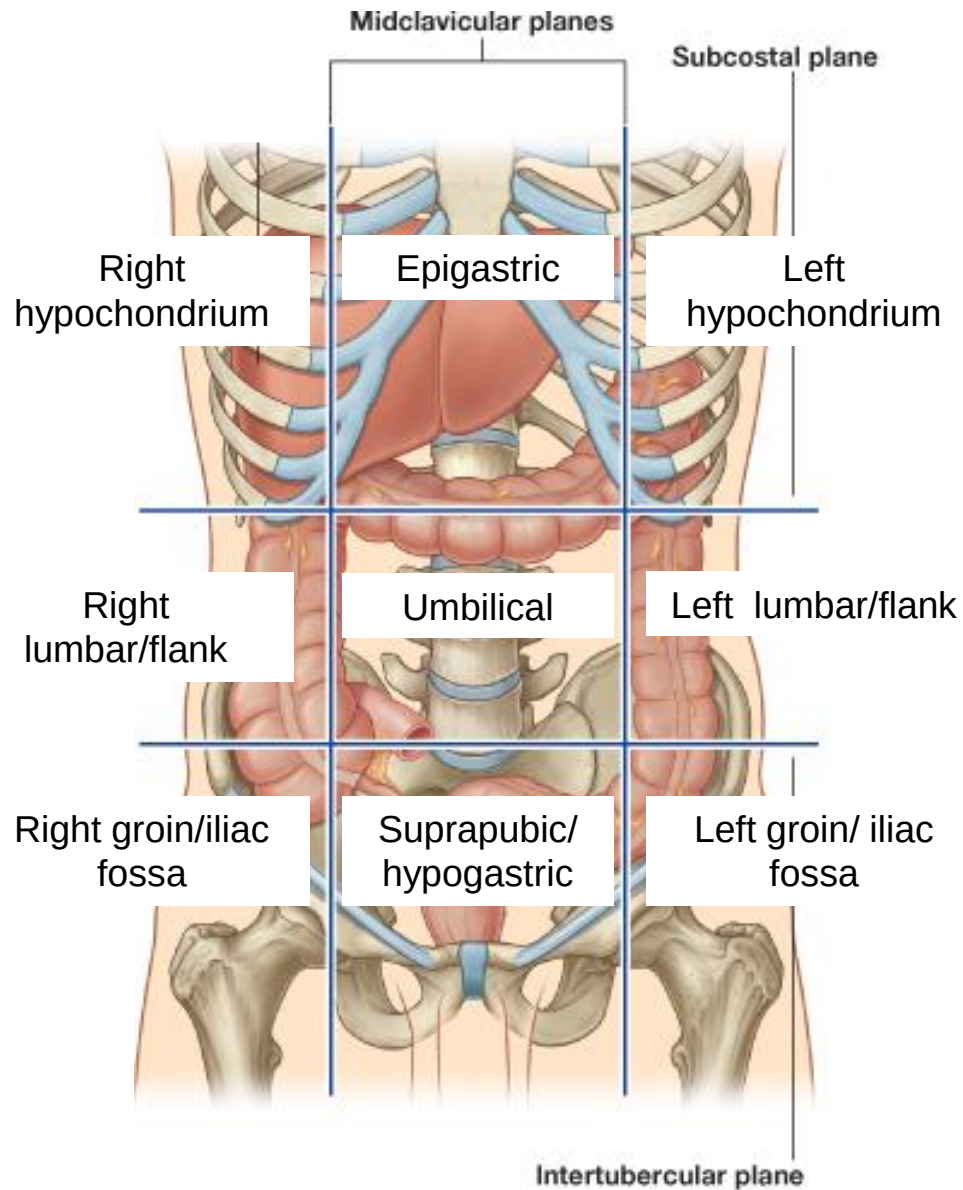
Referred Pain



- Sensory information enters the spinal cord from one location, but is interpreted by the CNS as coming from another location
- For example, the gut (visceral) have low sensory output and these sensory afferents converge on the same spinal cord level that receive sensory afferents from high output areas such as the body wall (somatic)
- As a result the low sensory output is interpreted as coming from the area of high sensory output
- Pain can also be referred from one somatic region to another.



Review



Referred Pain in the GI

Visceral afferents follow the autonomic supply “backwards” to the spinal cord

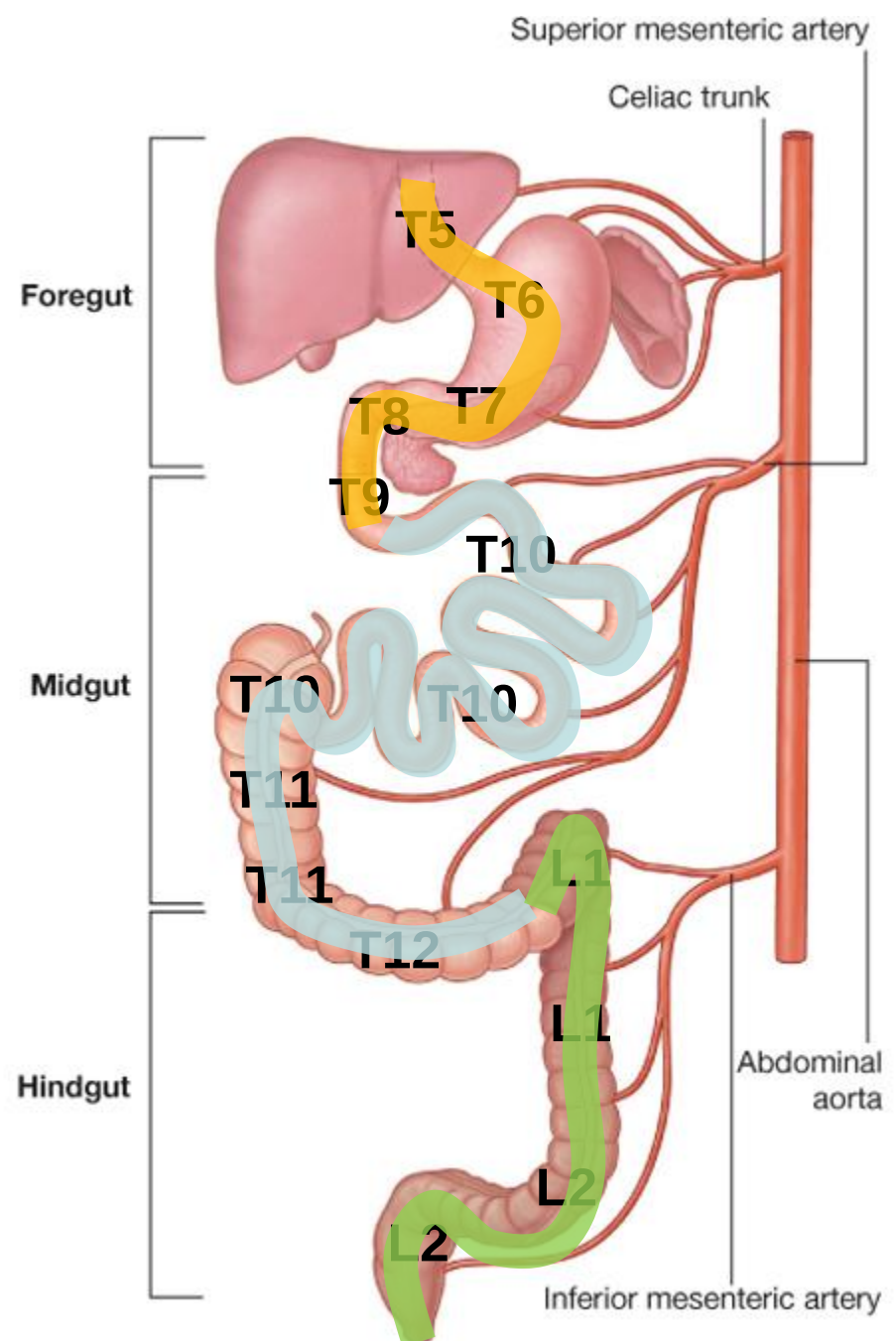
Generally, referred pain from →

Foregut Structures: perceived as epigastric/upper quadrant pain (sympathetics T5-9)

Midgut Structures: perceived as periumbilical pain (sympathetics T10-12)

Hindgut Structures (up to the midpoint of sigmoid colon): perceived as lower quadrant/ flank/ groin/ lower limb (sympathetics L1-2)

Hindgut Structures (after the midpoint of sigmoid colon): perceived as perineal pain (parasympathetics S2-S4)



Referred Pain in the GI

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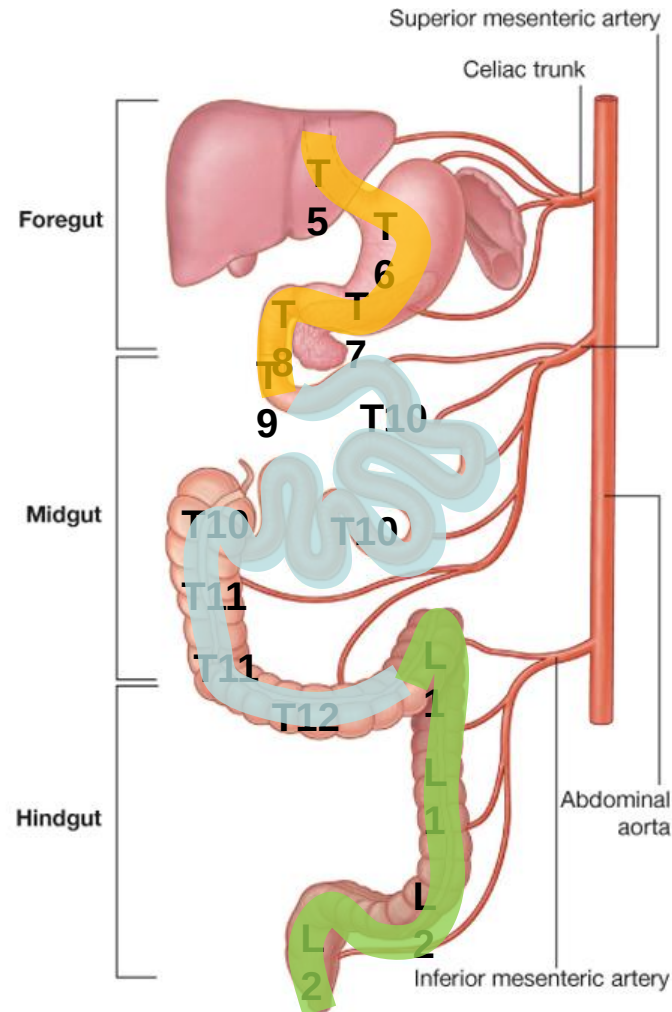
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GUT	DERIVATIVES	ARTERY	Preganglionic NERVE SUPPLY	
Foregut	ESOPHAGUS STOMACH PANCREAS LIVER GALL BLADDER BILE DUCT DUODENUM (cranial half)	CELIAC	Sympathetic	Parasympathetic
			GREATER THORACIC SPLANCHNIC NERVES	VAGUS
Midgut	DUODENUM (caudal half) JEJUNUM ILEUM CECUM & APPENDIX ASCENDING COLON TRANSVERSE COLON (proximal 2/3)	SUPERIOR MESENTERIC	LESSER and LEAST THORACIC SPLANCHNIC NERVES	
Hindgut	TRANSVERSE COLON (distal 1/3) DESCENDING COLON SIGMOID COLON RECTUM ANAL CANAL(cranial part)	INFERIOR MESENTERIC	LUMBAR SPLANCHNIC NERVES	PELVIC SPLANCHNIC NERVES